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Combining modelling and simulation approaches

How to measure performance of business processes

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Abstract

Purpose – The purpose of this paper is to provide a method for analysing and improving the operational performance of business processes (BPs).
Design/methodology/approach – The method employs two standards, Business Process Modelling Notation (BPMN 2.0) and Business Processes Simulation (BPSim 1.0), to measure key performance indicators (KPIs) of BPs and test for potential improvements. The BP is first modelled in BPMN 2.0. Operational performance can then be measured using BPSim 1.0. The process simulation also enables execution of reliable "what-if" analysis, allowing improvements of the actual processes under study. To confirm the validity of the method the authors provide an application to the healthcare domain, in which the authors conduct several simulation experiments. The case study examines a standardised patient arrival and treatment process in an orthopaedic-emergency room of a public hospital.
Findings – The method permits detection of process criticalities, as well as identifying the best corrective actions by means of the "what-if" analysis. The paper discusses both management and research implications of the method.
Originality/value – The study responds to current calls for holistic and sustainable approaches to business process management (BPM). It provides step-by-step process modelling and simulation that serve as a "virtual laboratory" to test potential improvements and verify their impact on operational performance, without the risk of error that would be involved in *ex-novo* simulation programming.

Keywords Performance, BPMN, Healthcare, Business process modelling, Business process simulation

Paper type Research paper

1. Introduction

Business process management (BPM) is a discipline dealing with the management of business processes (BPs) for the achievement of continuous improvement in organisations (vom Brocke *et al.*, 2014; Recker, 2014; Rosemann and vom Brocke, 2015). Organisations wishing to implement BPM must analyse the operational performance of the BPs in place, as well as design and model potential improvements (Aguilar-Savén, 2004; Dey *et al.*, 2006;

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McCormack *et al.*, 2009; Trkman, 2010). Operational performance analysis should be carried out using a holistic systems-based approach, applying both qualitative and quantitative performance criteria (Vernadat *et al.*, 2013; Rosemann and vom Brocke, 2015). The criteria chosen must be consistent with organisational objectives and provide cyclical support for organisational decision-making and strategy setting (Nudurupati *et al.*, 2011; Vallejo *et al.*, 2012).

Many researchers advocate the use of simulation for analysing BPs and their relative operational performance. BP simulation is in fact a widely used approach in BPM (van der Aalst *et al.*, 2003; Giaglis *et al.*, 2005; Wang *et al.*, 2009; Suzuki *et al.*, 2012), and is recognised as one of the most effective for understanding and correcting operational performance issues (Trkman, 2010). Discrete-event simulation is one of the most popular simulation techniques (Robinson, 2005). Businesses can use simulation to gain knowledge of a given process, particularly through "what-if" analysis, in which the outcomes of potential alternative process designs are compared through simulated test statistics (Pidd, 2009; Robinson, 2014).

The current paper is intended to respond to the identified need for holistic, sustainable BPM-oriented approaches (Rosemann and vom Brocke, 2015). It does so by proposing a method that combines both modelling and simulation of BPs, thus permitting the "virtual analysis" of organisational alternatives and strategies. The combination of modelling and simulation can assist in first visualising process behaviour, then measuring process operational performance, and finally examining the different scenarios for improvement (Pidd, 2009; Robinson, 2014). The proposed conceptual and quantitative method employs both the Business Process Modelling Notation (BPMN 2.0) modelling standard and the Business Processes Simulation (BPSim 1.0) simulation standard to enable measurement of operational performance under different BPs (Wohed *et al.*, 2006; Vergidis *et al.*, 2008), and thus to select appropriate processes for improvement in organisational performance.

BPMN is a modelling notation designed to be readily understood by all stakeholders and to permit application by both technical and non-technical users (OMG, 2011). Compared to other modelling languages, BPMN has the advantage of supporting the design of conceptual models that can immediately be converted into process models in executable languages (i.e. Business Process Execution Language, BPEL) (Onggo, 2009), and directly executed in software systems. The use of BPMN notation also permits a good representation of the conceptual model, which can be defined as "the description of the model that is to be developed" (Onggo, 2009; Robinson, 2014). Conceptual modelling involves developing an understanding of the problem situation, determining the objectives to be modelled and the preliminary design of the input, output and model content. For a comprehensive explanation of conceptual modelling, refer to the literature (Pidd, 2009; Robinson, 2014).

The current paper illustrates "where" and "when" the proposed "combined" method can be applied, as well as "what" it can be used to analyse. As shown in the case study, the method permits testing of process designs and quantified measurement of the variations in operational performance, through several key performance indicators (KPIs). To validate the approach, we conduct a set of simulation experiments.

The next section of the paper provides a review of the relevant literature concerning simulation, particularly its applications in organisational and management studies and BPM. Section 3 describes the proposed method and the specific steps for applying the analysis. Section 4 presents the case study, in the healthcare sector. The case study includes running a set of simulation experiments to test the practicality and provide empirical validation of the method. The conclusions to the paper point out the practical implications as well as some limitations of the method, and indicate topics for further research.

2. Literature review

Simulation is a third way of carrying out scientific research, alongside theoretical and empirical analyses, which are in turn based on deductive and inductive reasoning (Axelrod, 1997). Similar to deduction, simulation begins with explicit assumptions (input data) from which we derive results (output data). However, simulation also involves induction, in that the relationships among the observed data (variables) can be analysed by means of experiments (simulations) (Harrison *et al.*, 2007; Robinson, 2014). Simulation offers potential advantages in dealing with situations that are analytically intractable under deductive methods, and where inductive methods encounter problems of unavailable data. Simulation also enables researchers to design and run empirical experiments that would be difficult or costly to perform with real people or systems, thus gaining understanding of the real world and ultimately permitting improvement (Jahangirian *et al.*, 2010). Simulation is particularly useful when the system under examination is complex, and characterised by numerous interactions. Simulation permits flexible, modular forms of experimental design and thus systematic approaches to organisational and management research in complex contexts. In summary, simulation offers advantages of cost effectiveness, time savings, control over experimental conditions, simplicity, openness to intuitive approaches and the potential of variability in modelling (Pidd, 2009; Robinson, 2014).

Appropriately designed simulation provides outcomes that are open to measurement and analysis, and relevant and useful to the study objectives. The results from different iterations and multiple variations in inputs (what-if analysis) permit the analyst or manager to deepen their knowledge of the real system under study (Harrison *et al.*, 2007; Pidd, 2009; Robinson, 2014). Hence, simulation is a useful tool for identifying best practices and supporting decision-making (Vallejo *et al.*, 2012). It can also be used to design new processes for response to situations that have not yet developed, and to test different process conditions and scenarios prior to experimentation and implementation in the real world (Januszczak, 2011). However, simulation can also give rise to errors, particularly arising from faults in the original programming design or from incorrect generalisation of outcomes to situations where the conditions are different from the simulated scenario (Harrison *et al.*, 2007).

Prior to the 1990s, simulation-based approaches were applied in different avenues of scientific research, but most frequently in the social sciences (Harrison *et al.*, 2007). Researchers in organisational and management studies did not regularly use simulation, and tended to overlook its potential contribution to their work. It was only beginning in the 1990s that simulation approaches began to appear more frequently in the major management journals, and researchers in operations management are among those that show increasing interest in computer simulation (Pannirselvam *et al.*, 1999; Shafer and Smunt, 2004; Harrison *et al.*, 2007).

The simulation-based approach has also been a powerful tool in the study of BPs and operational performance. An extensive history of BPM literature (e.g. from Giaglis *et al.*, 2005 to vom Brocke *et al.*, 2014) demonstrates the effectiveness of BP simulation as an organisational research method. The simulation approach appears particularly beneficial when applied to the study of complex real-world systems and organisations, in combination with the other main approaches in BP management: namely BP modelling and performance measurement (Antunes and Mourão, 2011; Glykas, 2013). Indeed, BPM is a discipline where research should take a holistic approach (Rosemann and vom Brocke, 2015). The aim of BPM is to analyse BPs and their interactions, identifying potential improvements as a support to decision makers, in a manner that addresses the multidisciplinary nature and the complexity of the true processes (Margherita, 2014; Macedo de Morais *et al.*, 2014; Bolsinger *et al.*, 2015). BPM is recognised as requiring organisational capabilities that draw on knowledge from different fields, such as information technology, operations management, behavioural psychology and other sciences, in order to design, analyse and improve the BPs (Recker, 2014; vom Brocke *et al.*, 2014; Rosemann and vom Brocke, 2015). BPM techniques should be applicable to a wide range of domains, and be capable of dealing with rapid change in dynamic contexts (Suzuki *et al.*, 2012).

Currently, there is a high demand for practical approaches to BP analysis and improvement, yet the existing approaches are fragmented and inconsistently applied (Bolsinger *et al.*, 2015). BPM approaches should be consistent with the overall organisational strategy, and permit continuous improvement and management control over BPs and their performance (Recker, 2014; vom Brocke *et al.*, 2014; Rosemann and vom Brocke, 2015). In line with these considerations, the current study combines three of the major approaches of BPM, such as process modelling, simulation and performance measurement, in a single method for the measurement and improvement of BPs.

3. The method

This section presents the method for measuring and improving operational performance of BPs, based on application of the BPMN 2.0 modelling standard and the BPSim 1.0 simulation standard. The step-by-step modelling and simulation method can also be employed as a virtual laboratory, for what-if analysis and testing of potential improvements, verifying their impact on performance without the risk of error in *ex-novo* simulation programming. We refer to the proposed method as "simulation-based process performance analysis", or "SimPPA".

BPMN is a standard graphic notation language for BP modelling, defined and maintained by the Object Management Group (OMG). BPMN can be used for both stand-alone modelling of BPs and for developing web-based applications. The modelling process in BPMN is flexible and intuitive, and is readily understood by theoreticians, applied researchers, technical developers and business people. In organisational environments, BPMN permits designers, implementers and managers of BPs to fill the gaps between their competencies (OMG, 2011). BPSim is a standard for BP simulation defined by the Workflow Management Coalition. The standard provides specifications enabling the parameterisation and interchange of process analysis data, and thus the structural, capacity and performance analysis of process models. BPSim 1.0 enables both pre-execution and post-execution analysis of BPs (www.BPSim.org). Moreover, BPSim 1.0 allows implementation of simulation modelling, which is especially useful in examining organisational systems involving queues. In the current study we take advantage of these qualities to investigate just such a system, specifically a hospital emergency department.

Several types of software can be used to simultaneously manage both modelling in BPMN 2.0 and simulation using the BPSim 1.0 standard. For the current research we choose the Bizagi Process Modeller. We make this choice for several reasons. First because the company's software supports both BPMN 2.0 and BPSim 1.0, second because Bizagi is a freeware tool, and third because it is respected among editors for its ease of use and extensive validation (Chinosi and Trombetta, 2012). In fact Bizagi is one of the most active members in the OMG, and operators can use Bizagi software for building and simulating BPMN diagrams, without requirement of licence fees.

To begin our proposed process analysis method, the user must first identify the type of process under examination. For this we adopt a process matrix (Figure 1) based on an original design by Hall and Johnson (2009), for classifying the process types susceptible to analysis under the method (the "where" and "when" to apply SimPPA). The grid classifies processes according with two variables, "process environment variability" and "output-variation value to customers", thus identifying four typologies:

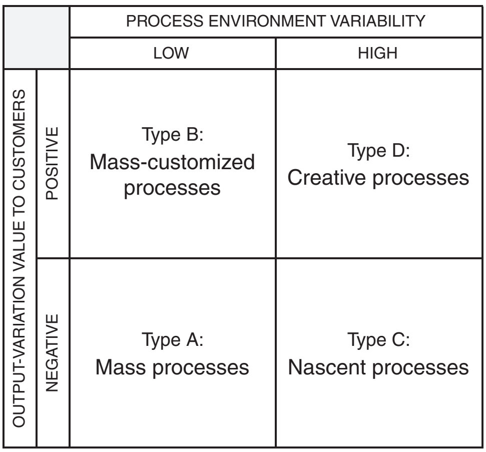
(1) mass processes are standardised processes characterised by no variations of output;

(2) mass-customised processes are standard processes or processes following a protocol to produce controlled variations in output;

(3) nascent processes are processes that do not yet exist or that are as yet "not controlled", involving first-time use of materials, technologies or designs; and

(4) creative processes are processes where customers demand and perceive variations in output.

The four types of processes must be managed according to their different characteristics. The SimPPA method can be applied to mass process (type A), provided that the complexity in the system is not because of randomness (i.e. variability in the process) but rather because of multi-actor involvement. In the case of mass-customised processes (type B), the element of standardisation permits continuous measurement and improvement of operational performance and thus application of the SimPPA method. For nascent processes (type C), an initial step is required in which the design is first formalised. At that point, BPs can be standardised and subject to SimPPA method, once again in support of continuous measurement and improvement of operational performance. Lastly, under creative processes (type D), the process environment variability is higher. In this case, continuous measurement and improvement of



Source: Hall and Johnson (2009)

Figure 1.
Matrix for process classification

operational performance can be misleading or less useful, and accordingly application of SimPPA method is not the best option.

Figure 2 summarises the steps in the SimPPA method, to be applied once the process type is identified.

The following sub-sections illustrate the main steps of the SimPPA method.

Define the variables for measuring operational performance: although many approaches are possible for measuring operational performance (Franco-Santos *et al.*, 2007; Franceschini *et al.*, 2013), the current research follows the precedent of van der Vaart *et al.* (2011), who stress the consideration of the various operational performance criteria through an integrated set of measures. The four criteria considered in this research are: completion rate of process, throughput time, rate of resource utilisation and resource service level. The criteria are defined as follows:

(1) completion rate of process is the ratio of completed instances to started instances;

(2) throughput time is the mean time for passing through the entire process;

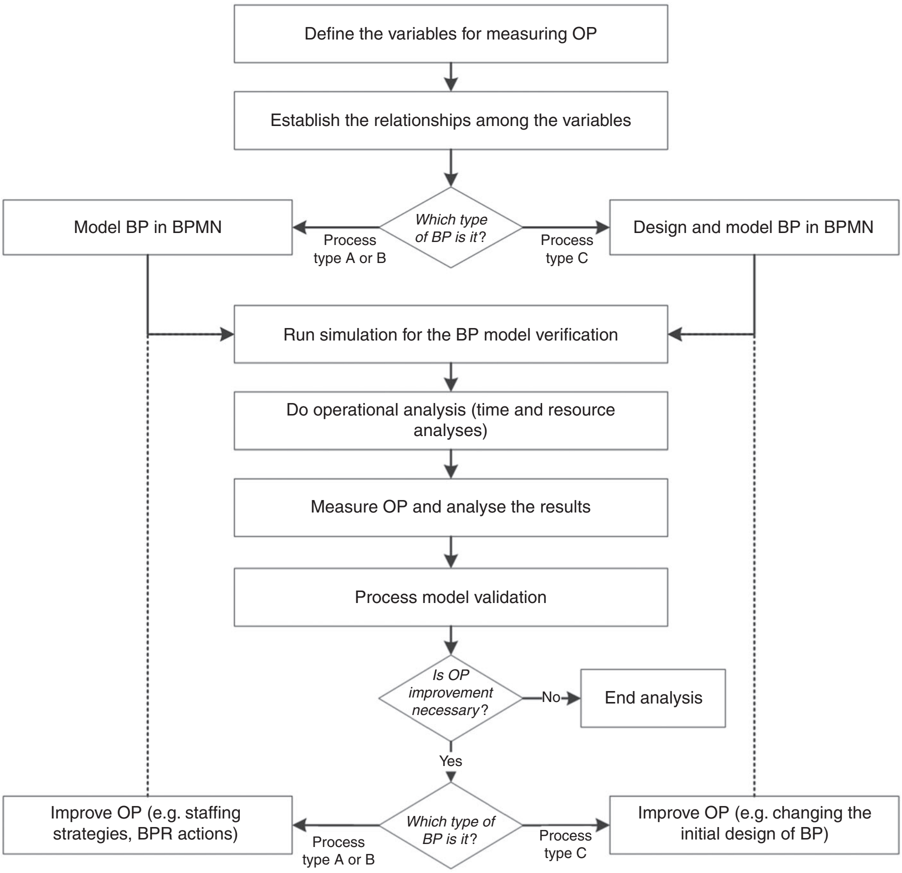


Figure 2. Main steps in a SimPPA method analysis

(3) rate of resource utilisation is the ratio between the effective and the maximal theoretical use of a resource (both human and technical); and

(4) resource service level measures the average waiting times of all tasks of the process due to the ith resource (as a proxy of service quality).

These criteria are effectively the "what" aspects of operational performance that the proposed method is intended to analyse. They are assessed through measuring the following variables: started (i_s) and completed instances (i_c), throughput time (ThT), resources utilisation of the process (u_{Ri}) and average waiting time for each resource of the process (AWT). These variables are organised in four KPIs, corresponding to the four operational performance criteria of interest, as shown in Table I.

In the following paragraphs we outline the steps in the method.

Establish the relationships among the variables: given started instances (i_s) and time (T_{sc}), the measurement of any BP operational performance becomes meaningful if and when the BP is capable of completing all started instances ($i_c = i_s$) in the available time (T_{sc}). Thus the necessary condition is:

$$ThT^* + \tau_a(i_s - 1) + AWT_{tot} \leqslant T_{sc} \tag{1}$$

where:

(1) ThT^* is the throughput time of the process when an infinite resource capacity is assumed. ThT^* is the best-case scenario and the benchmark value for the process, and is identified from the outcome of time analysis.

(2) τ_a is the time between arrivals after the first arrival ($i_s - 1$), occurring in the time T_{sc}. It has a Poisson distribution (as the best distribution of arrival times) and is shown in the following equation:

$$p_P(i_s)\big|_{T_{sc}} = e^{-\lambda T_{sc}} \frac{(\lambda T_{sc})^{i_s}}{i_s!} \tag{2}$$

where, λ is the arrival intensity or the expected number of arrivals occurring per each time unit (its inverse is a time, in particular $\overline{\tau}_a = 1/\lambda$, where $\overline{\tau}_a$ is the mean of τ_a); (λT_{sc})

Table I. Key performance indicators (KPIs) for measuring OP

OP criteria	OP variables	KPIs		KPIs domain
Completion rate of process	i_c: completed instances of the process i_s: started instances of the process	$KPI_1 = i_c/i_s$	KPI_1 is a measure of the completed process instances compared to started process instances	$0 \leqslant KPI_1 \leqslant 1$
Throughput time	ThT: process throughput time ThT^*: benchmark value for the process throughput time	$KPI_2 = 1 - \frac{ThT^*}{ThT}$	KPI_2 is a measure of the ThT variation compared to ThT^*	$0 \leqslant KPI_2 \leqslant 1$
Rate of resource utilisation	u_{Ri}: ith resource utilisation for the process	$KPI_3 = u_{R_i}$	KPI_3 is as a measure of the utilisation of a process resource	$0 \leqslant KPI_3 \leqslant 1$
Resource service level	$\sum_{Ri} AWT$: sum of the average waiting times for all the process tasks linked to the ith resource	$KPI_4 = \frac{\sum_{R_i} AWT}{ThT}$	KPI_4 is a measure of the delay caused by a process resource, relative to ThT	$0 \leqslant KPI_4 \leqslant 1$

is the mean number of events (i_s) occurred in the time T_{sc}; e is the base of the natural logarithm, approximately 2.718; $i_s!$ is the factorial of i_s. Accordingly, $p_p(i_s)|_{Tsc}$ represents the probability that events occur in a time T_{sc} with an arrival intensity of λ:

(1) AWT_{tot} is the total of average waiting times. It depends directly on resource utilisation (u_{Ri}) and consequently on the started instances (i_s), as seen in the formula below. For each ith resource:

$$AWT = f(u_{Ri}(i_s)) \tag{3}$$

where AWT is the average waiting times of all tasks of the process linked to the ith resource. Calculation of AWT_{tot} can be difficult when the process is complex, and in this case simulation will be helpful.

(2) T_{sc} is the scenario time, which is the fixed time available for operating a process (days, hours, minutes).

Design and/or model BP in BPMN: software such as the Bizagi application allows modelling of processes in a simple and intuitive way, in compliance with the BPMN 2.0 standard (Vergidis *et al.*, 2008; Chinosi and Trombetta, 2012). In particular, the BP diagram that is specified by BPMN is one of the main diagrams for representing the "conceptual model" (Onggo, 2009; Robinson, 2014).

Run simulation for the BP model verification: the simulation approach to process research facilitates management by allowing *ex-ante* performance analysis of potential changes in BP. Simulation can also be used in "what-if" analyses, to gain greater knowledge of the processes under study and assist in improving their organisation.

The simulation standard adopted in the SimPPA method is BPSim 1.0. BPSim 1.0 permits assessment of the performance of BPs modelled in BPMN 2.0 through dynamic analysis using discrete simulation methods. The dynamic form of analysis considers process variation over time, as well as being deterministic in its simulation of inputs and outputs (Robinson, 2005; Jahangirian *et al.*, 2010). The analysis considers processes as discrete sequences of events, in which an event variation can change the entire state of the process. Thus, it supports decision-making where managers wish to analyse the individual and joint effects of resource utilisation, queuing, capacity, material flow requirements and other factors on the overall process (e.g. Kumar and Nottestad, 2009).

A first run of the simulation is executed for the model verification to ensure that the programme represents the model properly (Pidd, 2009). The verification of the model tests whether the model works as expected and is reliable, meaning that the relations between inputs and outputs are the same of those of the model. The process model is correct if each instance can pass through the entire sequence of process flows and all the started instances are completed in a planned time. At the stage of process model verification, times and resources are not yet inserted in the software.

Carry out operational (time and resource) analysis and measure operational performance: the operational analysis consists of the time and resource analyses of the process under study. In order to perform these analyses, process characteristics (e.g. times, resources, information, rules) have to be collected by analysing the real process and subsequently entered in the simulation software (in this case Bizagi Process Modeller). Randomness can be taken in consideration by using probabilities for process flows and statistical distribution (i.e. Poisson distribution for interval arrival times or exponential distribution for activities process times) in order to reproduce unpredictability in process times of tasks.

The purpose of the time analysis is to measure throughput times in a manner that provides a benchmark value for the process (ThT^*). Calculation of ThT^* can be based either on the actual observed input times for the process activities or on set standards. This kind of analysis assumes infinite resource capacity, and as a consequence does not consider process delays.

In the resource analysis, the human and technical resources that characterise a process are identified and inserted in the software. At this point we can observe the effects of introducing resource constraints, for example in creating bottlenecks and waiting times, thus reducing the operational performance.

Process model validation: the model validation determines whether the input/output relation of the model is the same of that of the real system through combining the observations both from the model and from the real system. The validation is done by using statistical tests in order to compare the simulated outputs with the field-collected data (Pidd, 2009).

Improve operational performance: the ultimate goal of the method is to support improvement in operational performance by evaluating how potential strategic and operational changes would influence the process and its KPIs ("what-if" analysis). The joint modelling-simulation approach offers a virtual laboratory that allows verification of the impact of process improvements on operational performance. Simulation and what-if analysis permit ready consideration of different scenarios in order to verify which changes would be best, testing even those that might seem much less probable. Simulation makes it possible to anticipate the occurrence of a range of different scenarios by introducing different initial conditions, involving changes in resource availabilities, times or the implementation of BP re-engineering actions.

In considering potential improvements in operational performance, KPI_1 must always be held equal to one (Equation (1)). The other KPIs serve to identify process criticalities, and thus to choose most appropriate actions. However, it may well be the case that improving some performance indicators will worsen other indicators. Such trade-off in KPIs must be correctly identified and managed, and analysis of operational performance must always be consistent with process priorities and critical factors.

4. Case study

The SimPPA method is designed for measuring operational performance of highly complex or random processes and for supporting their managerial improvement. This section presents an "exploratory case study" (Yin, 2014), meaning one that tests the reliability and practicality of the proposed method in the real world and draws on data describing the real situation of a 500-bed public hospital in northern Italy. The process examined is a standard hospital-established protocol for "yellow code" patient arrivals and urgent care treatment in the orthopaedic emergency room (OER). This particular case is a "mass-customised process" (type B), because it is a highly complex process involving multiple interactions among different care operators, but with low variability in the process environment, which tends to reduce the variability of outputs. One of the main problems in the emergency room studied is overcrowding, which impacts on patient-waiting times and increases dissatisfaction. Such problems are quite common in emergency rooms, and they are thus under constant pressure to reduce process variability and increase performance, and so also reduce expenditure and meet growing demands (Zhao *et al.*, 2015). Previous researchers (e.g. Abo-Hamad and Arisha, 2013) have indicated that the causes of overcrowding and patient-waiting times can include limits on the resources in the emergency room (doctors, nurses) and inefficiencies in operations, such as delays after admissions and in lab and radiology tests. Healthcare also typically represents one of the fastest growing areas of public expenditure. Thus there are needs for improved efficiency and effectiveness, increased productivity and at the same time enhanced patient care (Rutberg *et al.*, 2015).

The case study is built on data provided by the Italian hospital. The first step in the method was to build the conceptual model of the OER process, after meeting doctors and nursing staff, consulting the clinical guidelines adopted by the hospital and walking through the pathway on the ground. Second, the bulk of the data were gathered by querying the IT database. Where data from the IT system were lacking, the researchers interviewed the staff to overcome the insufficiencies. The IT database provided the number and interval rates of the patients admitted, the throughput time (ThT) and the staff resource availability and shifts. The service times for each activity were estimated in consultation with the staff and through on-site observation of the process. After building and analysing the "as-is" process, we tested the results statistically. We then met again with the clinical staff to discuss the significance of the results from the "as-is" process and some potential scenarios of process improvement.

Define the variables

The variables identified for measuring operational performance in the OER process are: admitted (i_s) and discharged patients (i_c), throughput time (ThT), human and technical resource utilisation (u_{Ri}) and average wait time for each resource (AWT). In the SimPPA method, measurement of operational performance is conducted by observing the interrelated performance in terms of these four variables, as indicated by the KPIs (KPI_1 to KPI_4), and considering the priorities and critical factors identified for the process.

Establish the relationship among the variables

The relationships among the variables are structured as seen in Equation (1). In the case study, we ensure that the conditions of (1) always hold true, meaning that all started instances are completed in the specified time.

Model BP in BPMN

Based on the data collected, we developed a model of the clinical process in BPMN 2.0. It allows visualising BPs in a simple and supportive way. Such modelling of healthcare processes provides a welcome support to the decision makers involved (Proudlove *et al.*, 2007).

Figure 3 shows the process for "yellow-code" patients arriving in the OER. The diagram is constructed using the main BPMN 2.0 graphic elements, namely flow objects, connecting objects, swimlanes and data. Each graphical element provides specific information necessary for description of the process. The elements are first identified and then organised in logical sequence, reflecting their reciprocal relationships and constraints. For example the different clinical operators and hospital units are modelled in different pools or lanes, permitting separate analysis of the actors, units and their interactions. In fact BPMN diagramming permits unit-specific modelling (in this case for the OER) that accounts for all the connections between the elements of a single system with those of the other organisational units (e.g. radiology, lab, etc.). This is particularly important in modelling and simulating complex systems such as healthcare processes. It helps overcome problems typical of some other forms of unit-specific modelling, "which usually ignore what happens over the other side of the wall" (Gunal and Pidd, 2010).

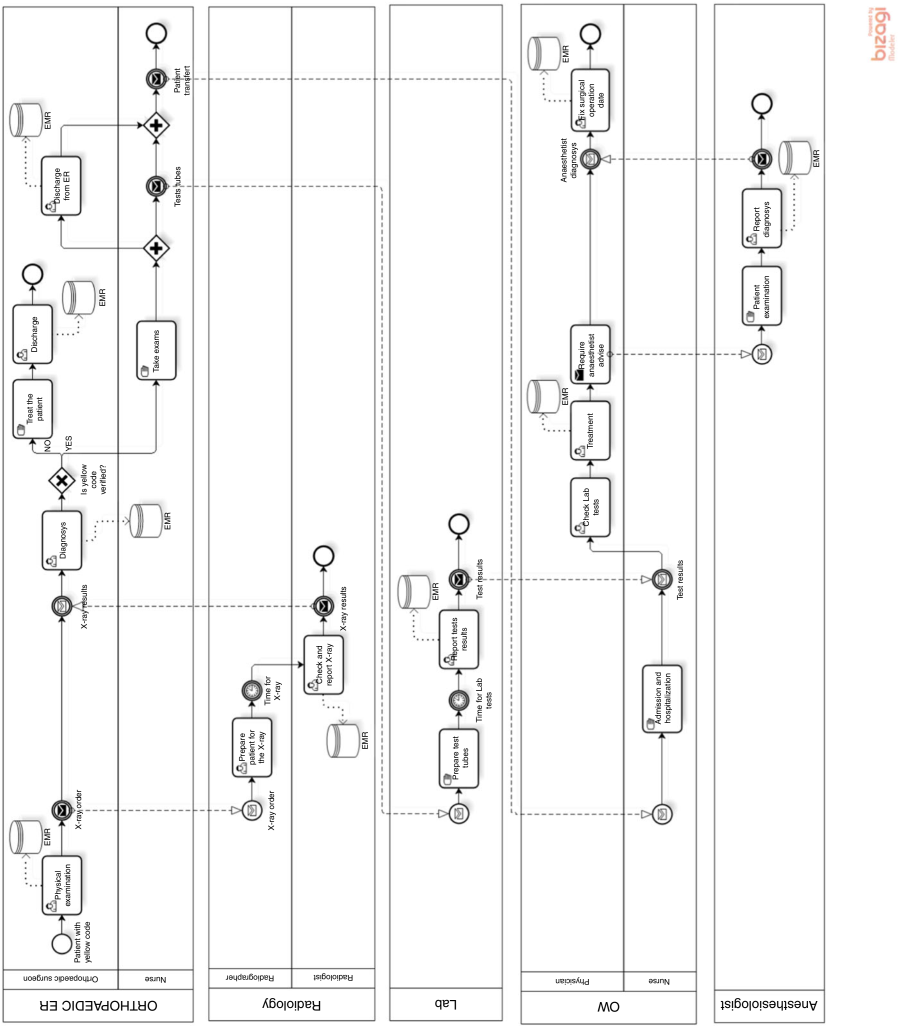


Figure 3.
Process of "yellow-code" patient arrivals and treatment in an orthopaedic emergency room (BPMN diagram)

Run simulation for the BP model verification
Once modelled, the software process was validated by verifying that all instances could be completed in six hours, corresponding to one morning shift. At this stage of the process model validation, times and resources were not yet inserted in the software. In keeping with the BPSim 1.0 standard, the process validation considered 1,000 instances (patient arrivals) in order to test software stability.

Do operational analysis
Following completion of the model validation, other data were entered into the software to carry out the operational analyses.

For the time analysis, the data required were the times of all individual activities necessary to solve a yellow code path, and the number of yellow code arrivals in the OER. The data show that the arrival time of patients has a Poisson distribution, meaning that the case is well suited to analysis using the SimPPA method (2), and the processing times are exponentially distributed. The analysis thus provided measurement of the benchmark process throughput time, *ThT**.

For the resource analysis, typologies, numbers and availability of both human and technical resources during the fixed time must be entered into the software. In the case under study, the OER morning shift provides for the availability of two nurses (resource 1) and one orthopaedic surgeon (resource 2). Therefore, a ratio of 2 to 1 between nurses and physician is considered. This ratio is in line with the data on the overall Italian healthcare system, but slightly less than for most other OECD countries (OECD, 2014). The analysis also considers the availability of radiology and laboratory services for support of specific OER activities. Inserting these standard resources in the model permits identification of occurrences of bottlenecks and waiting times, which reduce the operational performance in terms of time analysis.

Measure operational performance and analyse the results
Once the resource analysis is completed, the four KPIs are calculated, as presented in Figure 4.

To analyse operational performance we must first verify that all started instances are completed ($i_c = i_s$ and so $KPI_1 = 1$) in the available time (for this scenario, T_{sc} = six

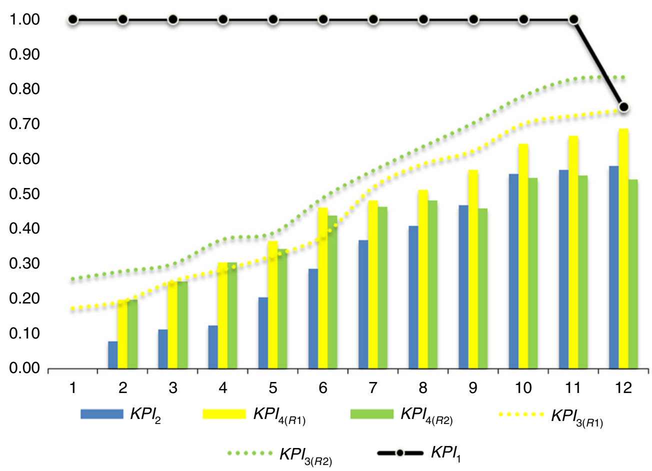


Figure 4.
Resource analysis results (KPIs)

hours). This step verifies the existence of the necessary condition expressed in Equation (1). As Figure 4 shows, KPI_1 is indeed equal to one, up to the case of eleven patient arrivals ($i_s = 11$). The condition expressed in Equation (1) is no longer valid when patient arrivals exceed eleven ($i_s > 11$). As a result, the SimPPA method calculates the maximum number of patients that the standard process considered can handle.

Once $KPI_1 = 1$ is established, we can then analyse the other KPIs. For the morning shift, the indicators show a growth trend with increasing patient arrivals. When the process operates at its maximum capacity ($i_s = 11$) we observe the following results: throughput time ThT increases compared to the benchmark value ThT* ($KPI_2 = 0.57$, i.e. ThT increases of 57 per cent); utilisation of OER nurses (resource 1) is lesser than utilisation of OER surgeon (resource 2) ($KPI_{3(R1)} = 0.72$, i.e., utilisation of resource 1 is 72 per cent and $KPI_{3(R2)} = 0.83$, i.e., utilisation of resource 2 is 83 per cent). Also, the impact on the overall delay time is greater from nurses (resource 1) than from the surgeon (resource 2) ($KPI_{4(R1)} = 0.67$, $KPI_{4(R2)} = 0.55$). These results show that when this highly complex and standardised healthcare process is used to its maximum production capacity, physicians are the most utilised resource but their impact on the overall delay time is less than the impact of the nurses.

Process model validation

After analysing operational performance through the "as-is" model, we must compare its simulated outputs with the field-collected data. Unfortunately, the data available for the case study hospital are concentrated on describing the throughput times of the OER process, concentrating on the time between "patient admission at the OER" and "discharge from OER". This situation is not unusual in healthcare (Santibáñez *et al.*, 2009). Indeed, most of the remaining data were obtained by interviewing the hospital staff and observing the process on the ground. We build 95 per cent confidential intervals (CIs) to evaluate the interval estimation of the actual data regarding the process throughput times. Then, we verify if the simulated process throughput time ($ThT = 121.67$ min) falls within the 95 per cent CI of collected data.

As shown in the Table II, the simulated process throughput time ($ThT = 121.67$ min) indeed falls within the 95 per cent CI of the actual data. Thus, the overall simulation model accurately represents the OER process of the hospital under study (although we could not perform validation for all the component performance indicators included in the analysis).

Improve operational performance

The proposed method offers a "virtual laboratory" for what-if analysis. In the following paragraphs we show how this laboratory can be used for testing the impact of potential improvements on performance.

Table II. Process model validation (95 per cent confidence intervals for actual ThTs)

	Mean	SE	(95% CI)	
Actual ThT	115.9756	12.46886	90.7751	141.1761

The process under study complies with a standard hospital-established protocol, where the design and sequence of activities cannot be modified. Even in such restricted situations, changes in staffing strategies can still offer potential for improvement in operational performance (Abo-Hamad and Arisha, 2013). Here, the potential scenarios could involve better allocation of critical resources, specifically the OER surgeon and OER nurses. The potential improvements are represented by two different "to-be" scenarios for the OER: an additional nurse (scenario 1) or an additional orthopaedic surgeon (scenario 2). (In this particular hospital, an additional surgeon could quite easily be transferred from the orthopaedic ward to the OER, without adding any extra resources to the overall hospital system.) Table III shows the *KPI* results calculated for the to-be scenarios of the two proposed improvements. These are derived from the same starting data as for the as-is scenario, for the situation of the maximum number of patients processed in a unit of time (i.e. 11 patients entering the clinical process, $i_s = 11$). Working from these calculations, Table III then also shows the operational performance variations (ΔKPIs) that could be derived from the different to-be scenarios (1 and 2), as compared to the as-is situation (scenario 0).

From Table III, we observe that condition (1) is verified in all the scenarios, meaning that $KPI_1 = 1$. The analysis also shows that a greater improvement in throughput time (KPI_2) can be gained by adding an extra orthopaedic surgeon (scenario 2, where the reduction of ThT is about 42 per cent) than by adding an extra nurse (scenario 1, where the reduction of ThT is about 12 per cent). However, the table also shows that between the two choices of adding resources, it is the addition of the orthopaedic surgeon that causes the greater decrease in the relative resource's utilisation (KPI_3). In fact in scenario 2, utilisation of the OER surgeon ($KPI_{3(R2)}$) drops 51 per cent with the addition of another surgeon (resource 2), while in scenario 1, the resource utilisation for OER nurse ($KPI_{3(R1)}$) decreases 33 per cent with the addition of another nurse (resource 1). Also, the service levels for the two resources of the OER nurse and the OER surgeon ($KPI_{4(R1)}$, $KPI_{4(R2)}$) decrease more by adding a second orthopaedic surgeon (scenario 2). We recall that KPI_4 is defined as "the measure of the delay caused by a process resource, relative to ThT" (see Section 3). Thus a decrease in this *KPI* implies real-world improvement. In particular, in the two scenarios, the decreases in KPI_4 for the addition of the OER nurse and the OER surgeon are respectively 20 and 38 per cent.

In conclusion, the addition of an extra orthopaedic surgeon (scenario 2) is the choice that results in the greatest improvement in throughput time (KPI_2) and resource service level (KPI_4), achieving results that are better than those calculated not only for scenario 0, but also for scenario 1.

The SimPPA method permits identification and management of trade-offs in the KPIs, in keeping with the priorities and critical factors defined for the process. To improve operational performance, KPI_1 must always be maintained equal to one

Table III. KPIs under two proposed improvements ($i_s = 11$) and variations of KPIs (ΔKPIs) compared to scenario 0

Scenario	Improvement	KPI_1	KPI_2	$KPI_{3(R1)}$	$KPI_{3(R2)}$	$KPI_{4(R1)}$	$KPI_{4(R2)}$
Scenario 0 (sc0)	–	1.00	0.57	0.72	0.83	0.67	0.55
Scenario 1 (sc1)	1 more nurse	1.00	0.50 (–12%)	0.48 (–33%)	0.83	0.62 (–7%)	0.59 (+7%)
Scenario 2 (sc2)	1 more surgeon	1.00	0.33 (–42%)	0.72	0.41 (–51%)	0.47 (–20%)	0.34 (–38%)

(Equation (1)). The other KPIs serve in the identification of process criticalities, and thereby the choice of appropriate actions for improvement. However, improving some performance indicators (in this case KPI_2, KPI_4) may worsen other indicators (in this case KPI_3). For the standard process under examination, if the process priority is the reduction of waiting times (KPI_4) and consequently the throughput time of the process (KPI_2), the best improvement is that obtained by adding an orthopaedic surgeon (scenario 2). In fact the reductions of KPI_2 and KPI_4 are both greater in scenario 2 than in scenario 1. To further reveal the differences between scenario 0 and 2 we provide a trend analysis of all the KPIs, as seen in Figure 5. Compared to the present case (scenario 0), we observe that adding one more orthopaedic surgeon would increase the number of patients treated in the OER by 20 per cent, and at the same time reduce waiting times and process throughput times relative to the as-is situation (see KPI_2 and KPI_4).

5. Conclusions

The current study responds to the identified need for holistic and sustainable approaches in BPM (vom Brocke *et al.*, 2014; Recker, 2014; Rosemann and vom Brocke, 2015), through a method involving step-by-step application of process modelling and simulation. The method (called SimPPA) features an application of the BPMN 2.0 modelling standard and the BPSim 1.0 simulation standard, for measurement of specific performance indicators (KPIs). It also provides a "virtual laboratory" for testing of potential process improvements, verifying their impact on operational performance without the risk of error involved in *ex-novo* simulation programming. The SimPPA method offers useful techniques and tools for decision makers who wish to better understand and improve process performance, and is especially useful in cases of highly complex or random processes. In particular, the method begins with a matrix for

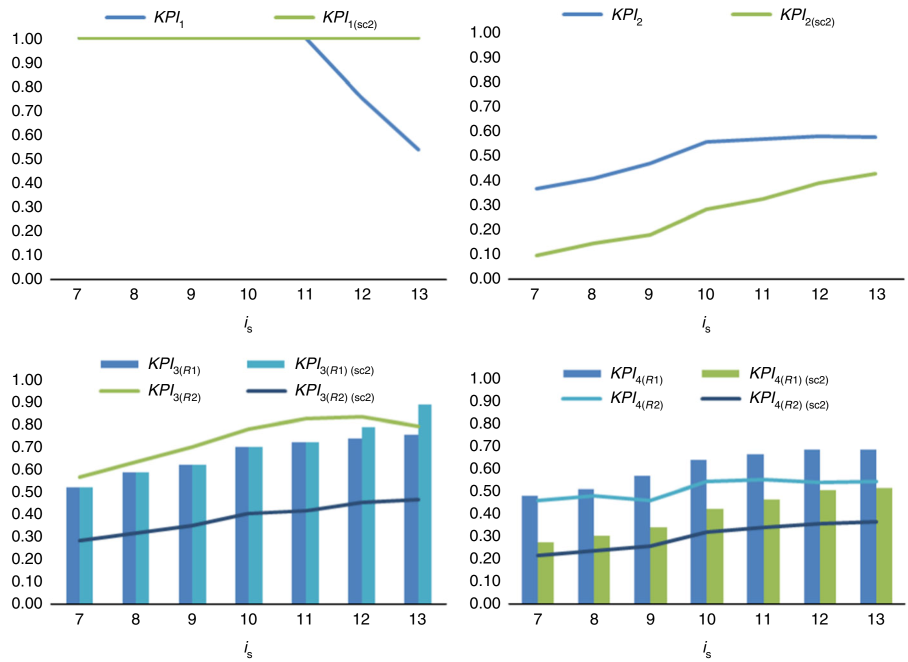


Figure 5.
KPIs of scenario 2 in comparison to the as-is case (scenario 0)

classifying the different organisational processes in terms of "process environment variability" and "output-variation value to customers". The subsequent methodological steps can then be satisfactorily designed to respond to a wide range of different processes types (Suzuki *et al.*, 2012).

This research demonstrates the effectiveness of BPMN for building a "conceptual model" that can be understood by all stakeholders (Onggo, 2009; Robinson, 2014). In fact, BPMN is a modelling notation specifically designed for ready understanding by all stakeholders and for application by both technical and non-technical users (OMG, 2011). Compared to other modelling languages, BPMN has the advantage of supporting the design of conceptual models that can immediately be converted into process models in executable languages (i.e. BPEL) (Onggo, 2012), and directly executed in software systems. Moreover, BPMN permits modelling of interactions between different actors in the process (in the study case, different hospital wards) and helps to overcome the problem typical of other unit-specific modelling, which generally fails to account for what is happening around the simulated system (Gunal and Pidd, 2010).

The research method responds to the real-life managerial difficulties in defining and achieving operational performance improvement for complex processes. To deal with this complexity, the proposed method adopts a systemic process-based approach for the analysis of operational performance, which searches for the potential trade-offs among several performance indicators (KPIs). As is seen from the case study outcomes, improving some performance indicators (in this case KPI_2 and KPI_4) may worsen other indicators (KPI_3). From a practical point of view, the SimPPA method can be considered as a roadmap for examining and carrying out BPM practices. The BPMN standard language is readily understandable by non-technicians, and so helps to bridge the gaps between different groups involved. As vom Brocke *et al.* (2014) stress, achievement of this "principle of joint understanding" is an essential feature of successful BPM. The various stakeholders should always be able to understand and analyse their organisational processes using a common language.

The current study is a preliminary effort, and the results require further refinement. The hospital unit examined shows characteristics that are typical of other orthopaedic-emergency units; however, the use of a single-case design has some limitations, precisely for issues of external validity and generalisation (Yin, 2014). Future studies could consider the replication of the research approach in different wards or hospitals, and also to combine further approaches (such as action research) to take account of more stakeholder engagement. A further limitation is that the current representation of performance criteria does not include financial costs among the indicators. Future research could extend the operational performance analysis to this and other performance dimensions in order to expand the range of decision-making input available to the manager. Other suitable topics for future research would be to compare the SimPPA method to other methods, and to combine it with other methods to provide new contributions to the analysis of operational performance of BPs.

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